Modeling and Control of Manipulators

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Enrico Simetti
DIBRIS - University of Genova
ISME - Interuniversity Research Center on Integrated Systems for Marine Environment
Via Opera Pia 13, Genova, Italy
enrico.simetti@unige.it

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Chapter 1

Introduction

Robotic manipulators, with their versatility and precision, have found their place in a wide range of applications across various industries and sectors. Here, we explore the diverse realms where robotic manipulators have made a significant impact:

- Manufacturing and Industrial Automation: Robotic arms are deployed in factories for tasks like welding, painting, and assembling. They enhance productivity, reduce error rates, and ensure consistent quality, revolutionizing manufacturing.
- Healthcare and Medical Robotics: Surgical robots assist surgeons in delicate procedures, reducing invasiveness and enabling remote surgeries.
- Space Exploration: Robotic arms on spacecraft and rovers collect samples, deploy instruments, and perform repairs in space and on other celestial bodies.
- Underwater Exploration: Robotic arms are routinely used in Remotely Operated Vehicles, to perform maintenance and repair operations in deep-water sites, where the presence of human divers is just not possible.
- Agriculture and Agribotics: Autonomous drones with manipulators perform planting, harvesting, and crop monitoring, increasing yields and sustainability in farming.
- Elderly assistance: There is an increasing trend in the adoption of human-like robots for the assistance of elderly people in day to day living, or just to keep them company.
- Entertainment and Creative Industries: Animatronic characters and precision-controlled camera rigs enhance entertainment experiences in theme parks and the film industry.
- Research and Development: Robotic manipulators are essential tools for testing prototypes, exploring materials, and developing new technologies.

In each of these applications, robotic manipulators contribute to increased efficiency, precision, and safety, pushing the boundaries of what technology can achieve.

1.1 Physics of a Manipulator

A manipulator is a mechanical device or robotic system designed to perform precise and controlled movements, often resembling the dexterity and range of motion of the human arm. In its essence, a manipulator is a serial connection of rigid bodies via rotational or translational objects, which allow the structure to move, even if such movements are limited. The rigid bodies are called *links* and the connection elements are called *joints*. Figure 1.1 shows the sketch of a typical 6 degrees of freedom (DOF) manipulator used in industrial settings. The DOF of a mechanical system are the number of independent parameters that define its configuration. In particular, the robot is composed by three 1-DOF rotational joints, and a 3-DOF ball wrist. For modeling purposes, the wrist can be seen as three 1-DOF rotational joints, whose origin is coinciding in the same point.

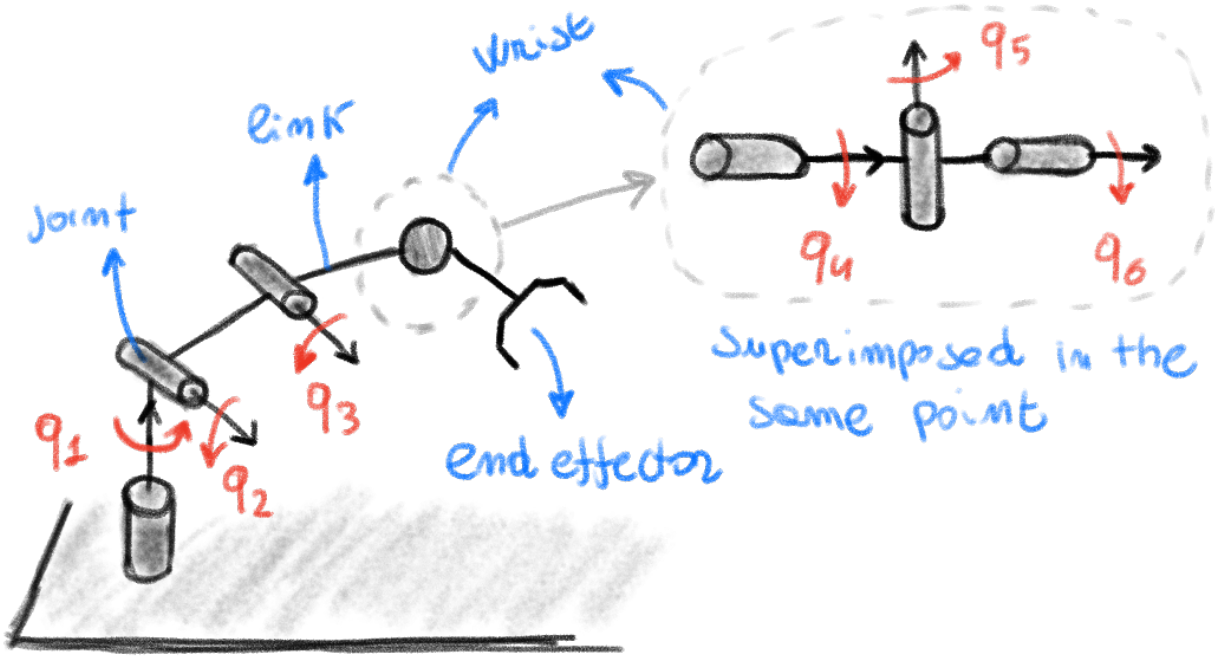


Figure 1.1: A typical 6 degrees of freedom manipulator

At a certain instant, the configuration of the robot is established by the values assigned to the various joints and mathematically is represented by the *joint configuration vector*

$$\boldsymbol{q} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix}, \tag{1.1}$$

where n is the number of joints and q_i represents the angle of rotation (for rotational joints) or the distance (for translational joints) with respect to a zero position initially established.

The example reported in the figure shows the structure of a manipulator with 6-DOF. There can be manipulators with less or more DOF, although that is the prevalent one employed in the industries. Indeed, having a structure with 6-DOF is the necessary condition to be able to locate the terminal body of the robot, often called the end-effector, in all possible orientations within a certain region of the 3 dimensional space. This region is called the *task workspace* of the robot.

As a comparison, the human arm has 7 DOF: 3 in the wrist, 1 in the elbow, and 3 in the shoulder. Having more than 6-DOF is useful to manage to robot configuration: the configuration can be changed without changing the position and orientation of the end-effector. Therefore, the 7-DOF configuration is *redundant* with respect the objective of keeping the end-effector in a given position and orientation.

The configuration of the robot changes over time. This depends on the rate of change of its joints, and is represented by the joint velocity vector

$$\dot{\boldsymbol{q}} = \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_n \end{bmatrix}. \tag{1.2}$$

In turn, the joint velocity vector changes in time due to the second derivative of the joint values, i.e.

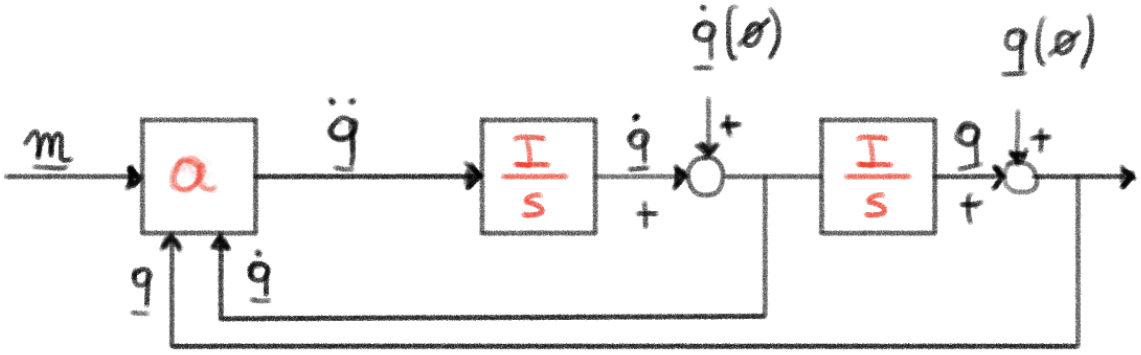


Figure 1.2: A closed loop scheme representing the dynamic model of a manipulator. Block "a" corresponds to equation (1.6).

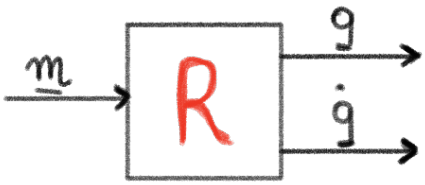


Figure 1.3: A compact representation of equation (1.6) and the integration of joint acceleration to obtain joint velocity and position vectors.

due to the joint acceleration vector

$$\ddot{\boldsymbol{q}} = \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \\ \vdots \\ \ddot{q}_n \end{bmatrix}. \tag{1.3}$$

There is no need to consider further derivatives, since the robot is an articulated mechanical system and its motion is originated from the torques and forces provided by electric motors. These values are organized in the so called join torque vector

$$\boldsymbol{m} = \begin{bmatrix} m_1 \\ m_2 \\ \vdots \\ m_n \end{bmatrix}. \tag{1.4}$$

The link between the torque provided by the motors, and the resulting joint acceleration is given by the dynamics of the manipulator. Mathematically, the dynamics can be modelled as

$$\boldsymbol{A}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{B}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q}) = \boldsymbol{m}, \tag{1.5}$$

where $\boldsymbol{A}(\boldsymbol{q}) \in \mathbb{R}^{n\times n}$ is the Inertia matrix, $\boldsymbol{B}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^{n\times n}$ is the Coriolis' effect matrix, and $\boldsymbol{c}(\boldsymbol{q})$ is a vector that keeps into account gravitational effects. In general, external forces different from gravitational forces can also be added on the side of $\boldsymbol{m}$ (for example, a force generated by a contact with the end-effector). Since $\boldsymbol{A}(\boldsymbol{q})$ is always invertible, the acceleration can be obtained as

$$\ddot{\boldsymbol{q}} = \boldsymbol{A}(\boldsymbol{q})^{-1}(\boldsymbol{m} - \boldsymbol{B}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{c}(\boldsymbol{q}). \tag{1.6}$$

The previous relationship will be more compactly represented as in Fig. 1.3. Clearly, if one generates $\boldsymbol{m}$ in some way, the robot will move, but the movement will not be predictable. As we want the robot to do a given task, we need to find out how to generate the torque vector appropriately.

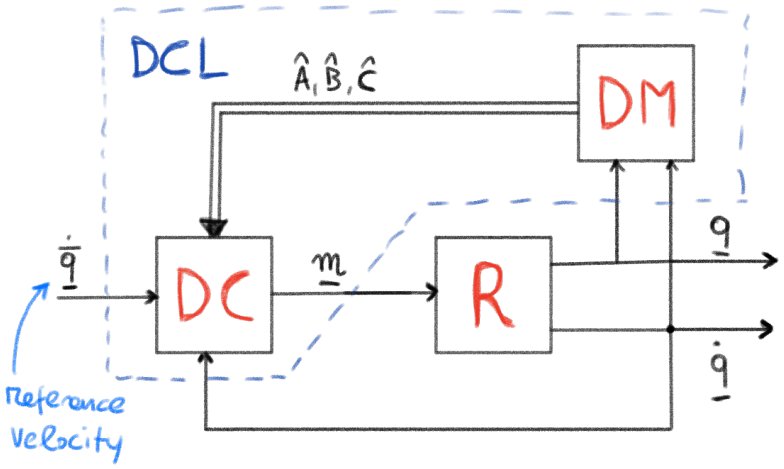


Figure 1.4: A block representation of the Dynamic Control Layer. The DM block provides the estimated matrices to the Dynamic Controller (DC) block, which generates the torque vector in input to the Robot.

1.2 A General Approach to Robot Control

We will consider a backstepping approach to the control of the manipulator. We will do so by organizing the functionalities and algorithms into an hierarchy of two layers of control.

1.2.1 Dynamic Control Layer

The first layer, called Dynamic Control Layer (DCL), is directly connected to the robot. It is in charge of making the robot commandable through joint velocity reference signals, instead of torques. To do so, it is necessary to design a controller that receives desired joint velocities as input, together with the feedback velocities, and generates a joint torque vector capable of driving the actual joint velocities toward the desired ones.

Different types of controllers can be build. For example, a feedback linearization controller, a sliding mode controller, or a simpler proportional integrative derivative (PID) controller. Let us consider a feedback linearization controller. To build such a controller, let us consider first a dynamic model functional block, which is an algorithm capable of estimating the matrices $\hat{\boldsymbol{A}}(\boldsymbol{q})$, $\hat{\boldsymbol{B}}(\boldsymbol{q}, \dot{\boldsymbol{q}})$, and $\hat{\boldsymbol{c}}(\boldsymbol{q})$ from the values of $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$. There are fast algorithms that provide the numerical values of these matrices and vector as output, with timings suitable for a real time implementation. On the basis of the availability of such matrices, a controller that can best approximate the desired velocity can be constructed. The resulting control scheme can be seen in 1.4. The controller is implemented in the Dynamic Controller (DC) block. It takes as reference the desired joint velocities $\dot{\underline{\boldsymbol{q}}} \in \mathbb{R}^n$, and as feedback the actual velocities $\dot{\boldsymbol{q}}$. Furthermore, it exploits the Dynamic Model block to have the estimated matrices of the dynamic model of the robot. The output is the joint torque vector $\boldsymbol{m}$, which is fed to the actuators of the robot.

If the performance of the controller are sufficiently good, and the references are slowly varying (i.e., they are inside the bandwidth of the controller), then the above representation can be approximated with the one shown in Fig. 1.4.

1.2.2 Kinematic Control Layer

On top of the DCL lies the Kinematic Control Layer (KCL). This layer is in charge of providing the desired reference velocity $\dot{\underline{\boldsymbol{q}}}$ to the underlying DCL block. This reference should be constructed so that the robot moves as desired in the Cartesian space.

To this aim, it is first necessary to understand how the arm is positioned in the Cartesian space. Intuitively, the positioning of each link is a function of the joint position vector $\boldsymbol{q}$, as the position and orientation of a link will depend on how the previous joints are rotated (or translated). To better

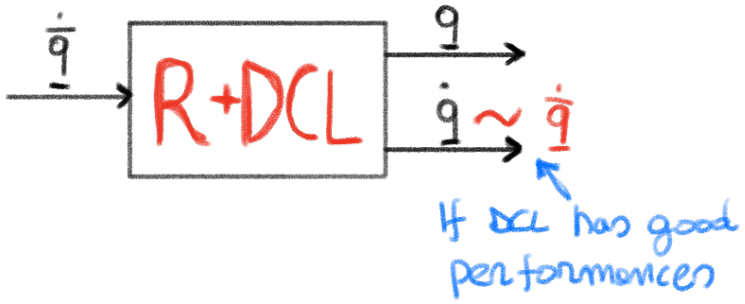


Figure 1.5: A block representation of the Dynamic Control Layer together with the Robot, forming a sort of Kinematic system. The output velocity can be approximated with the input desired velocity, if the performance of the DCL are sufficiently good.

understand the position and orientation of the links in the Cartesian space, let us attach a *frame* to each link, including to the base and to the end-effector. A frame is defined by three unitary vectors, orthogonal to each other, originating in the same point, which is called the origin of the frame.

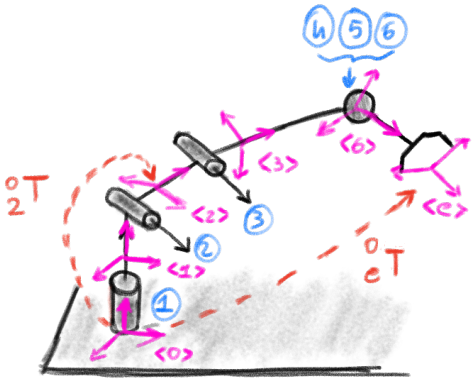


Figure 1.6: .

The mutual positioning and orientation of a frame with respect to another one is represented by a 4×4 matrix called *Transformation Matrix*. If one wants to indicate the positioning of the end-effector frame $\langle e \rangle$ with respect to the base frame $\langle 0 \rangle$, then one would write ${}^0_e\boldsymbol{T}(\boldsymbol{q})$.

Of course, a transformation matrix can be computed for any couple of frames. Typically, it is of interest to know the positioning of the robot frames with respect to the base frame, hence from the knowledge of the joint position vector $\boldsymbol{q}$ one can compute ${}^0_1\boldsymbol{T}(\boldsymbol{q}); {}^0_2\boldsymbol{T}(\boldsymbol{q}), \ldots, {}^0_n\boldsymbol{T}(\boldsymbol{q}), {}^0_e\boldsymbol{T}(\boldsymbol{q})$.

The knowledge of all these transformation matrices is the minimal amount of information to know the position and orientation of the different links of the robot. Hence, we can imagine to have a *Geometric Model* block (GM) that, knowing the physics of the robot and the current posture through the joint position vector $\boldsymbol{q}$, produces as output the transformation matrices mentioned above.

Now that we have a mean to compute the position and orientation of all the links in real-time, it is necessary to understand how the joints should move so that the end-effector reaches the desired position and orientation. In other words, it is necessary to understand how to make ${}^0_e\boldsymbol{T}(\boldsymbol{q})$ equal to a value that corresponds to the desired position and orientation.

Given the dependency of ${}^0_e\boldsymbol{T}$ on $\boldsymbol{q}$, it is necessary to act upon the joint velocity vector. As we are considering a backstepping approach, we need to generate an appropriate desired joint velocity vector $\dot{\boldsymbol{q}}$.

With reference to Fig. 1.7, let us consider a generic link $\langle i \rangle$. Such a frame changes its position and orientation over time, as the joints move. Therefore, it is characterized by a certain Cartesian velocity. Let us label such a velocity as $\dot{\boldsymbol{x}}_i \in \mathbb{R}^6$. This 6 dimensional vector is composed by two 3 dimensional vectors defined in the Cartesian space, nameley the angular velocity vector $\boldsymbol{\omega}_i$ and $\boldsymbol{v}_i$. These geometric vectors assume different values if we consider them on one frame or another one. To differentiate between these different instances of the same geometric vector, we use the notation ${}^0\boldsymbol{\omega}_i$, which considers the geometric vector projected into frame $\langle 0 \rangle$.

The vector $\dot{\boldsymbol{x}}_i$ is linearly related with the joint velocity vector through a posture-dependent matrix called the *Jacobian* matrix, leading to the following equation

$$\dot{\boldsymbol{x}}_i = \boldsymbol{J}_{i/0}(\boldsymbol{q})\dot{\boldsymbol{q}}, \tag{1.7}$$

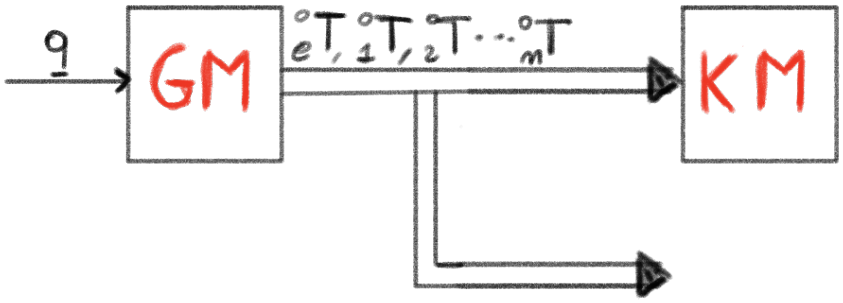


Figure 1.8: A block representation of.

where $\boldsymbol{J}_{i/0}(\boldsymbol{q}) \in \mathbb{R}^{6\times n}$ is the Jacobian matrix of frame $\langle i\rangle$ with respect to frame $\langle 0\rangle$. Given the serial nature of the manipulator, for a given frame $\langle i\rangle$, the only part of $\boldsymbol{q}$ that influence its velocity are the joints before and including joint i itself.

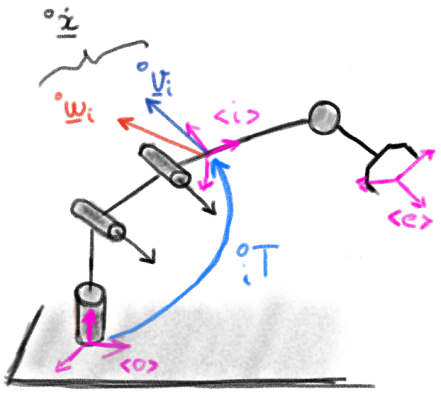


Figure 1.7: .

Hence, the previous relationship could also be written as

$$\dot{\boldsymbol{x}}_i = \boldsymbol{J}_{i/0}(q_1, \ldots, q_{i-1}) \begin{bmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_i \end{bmatrix}. \tag{1.8}$$

To evaluate this matrix for all the frames of the robot, it is important to highlight that the dependency of the Jacobian matrix on $\boldsymbol{q}$ is related to the position and orientation of all the frames preceding $\langle i\rangle$. Hence, it could be rewritten in terms of the transformation matrices of the preceding joints, as

$$\boldsymbol{J}_{i/0}(\boldsymbol{q}) = \boldsymbol{J}_{i/0}\left({}^0_1\boldsymbol{T}(\boldsymbol{q}), {}^0_2\boldsymbol{T}(\boldsymbol{q}), \ldots, {}^0_{i-1}\boldsymbol{T}(\boldsymbol{q})\right). \tag{1.9}$$

Therefore, it possible to construct a Kinematic Model functional block that takes as input the output of the Geometric Model block (the transformation matrices), and computes in real-time all the *basic Jacobians* of the manipulator as depicted in Fig. 1.8. The dependency on $\boldsymbol{q}$ is implicit in the dependency on the transformation matrices.

Let us now more formally consider the objective of the KCL, by tackling the most common case in robotics. This corresponds to the sketch drawn in Fig. 1.9, where the end-effector frame $\langle e\rangle$ needs to be driven to a desired *goal* position represented by the goal frame $\langle g\rangle$.

First, let us immediately note that the end-effector frame, like any other frame, can only be driven to a point that belongs to its workspace. Indeed, the movement of the frame is constrained by the manipulator's structure. However, if it were free from constraints, then we could easily find a velocity in the Cartesian space to move it towards $\langle g\rangle$, both in position and orientation. Hence, let us consider a functional block called *Cartesian Controller* (CC), which on the basis of the feedback position of the end-effector, represented by the transformation matrix ${}^0_e\boldsymbol{T}(\boldsymbol{q})$, and on the desired goal position ${}^0_g\boldsymbol{T}(\boldsymbol{q})$, generates the desired generalized velocity $\dot{\tilde{\boldsymbol{x}}}_e$ for the end-effector.

However, the frame is not free from constraints, but as it is attached to a manipulator's part, hence its movements depend on the motion of all the joints of the system. Therefore, this ideal Cartesian velocity must be attained with a specific motion of the joints, if possible. So a new fundamental block must be introduced to perform this mapping from the Cartesian velocity space to the joint velocity space. Such a block is called the *Inverse Kinematics* block and takes as input the desired Cartesian velocity $\dot{\tilde{\boldsymbol{x}}}_e$ and the Jacobian $\boldsymbol{J}_{e/0}(\boldsymbol{q})$ of the end-effector. The block provides the desired joint velocities $\dot{\boldsymbol{q}}$ that the DCL must track. These velocities, when applied, provide the *best approximation* to the desired Cartesian velocity. As an example, is the manipulator is constrained to move on planar surface,

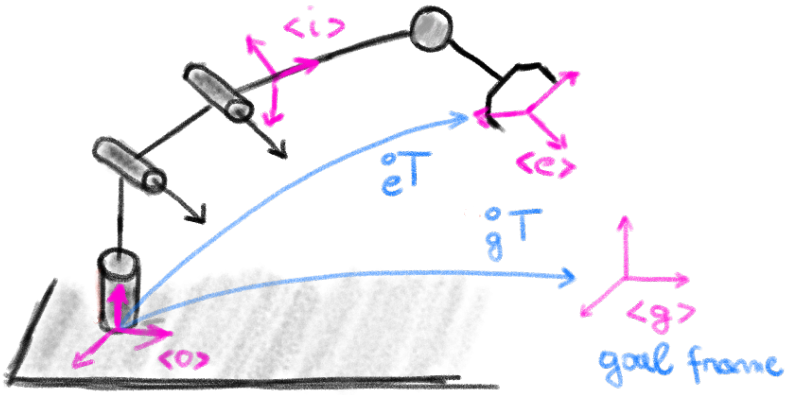


Figure 1.9: A block representation of.

and the goal is to catch an object flying in 3D space, then clearly the desired velocity $\dot{\boldsymbol{x}}_e$ will, in general, be outside of the plane and thus unattainable. The best approximation in this case will be reached by providing, in the Cartesian space, the projection of such desired velocity on the plane.

The overall control scheme can be seen in Fig. 1.10. The geometric model (GM), exploiting the joint position feedback, provides the transformation matrices to both the Kinematic Model (KM) and the Cartesian Controller. The Cartesian Controller, knowing the desired (goal) position for the end-effector represented by ${}^0_e\boldsymbol{T}(\boldsymbol{q})$, computes the desired Cartesian velocity $\dot{\tilde{\boldsymbol{x}}}_e$. This velocity is given as input to the Inverse Kinematics (IK) block, which finds the desired joint velocity vector $\dot{\boldsymbol{q}}$ that provides the best approximation. This velocity is the final output of the overall KCL and is given as input to the DCL, in particular to the Dynamic Controller (DC) block. In this particular example, the DC employs a feedback linearization approach, and hence receives the estimated dynamic parameters from the Dynamic Model (DM) block. On such a basis, it computes the joint wrench vector $\boldsymbol{m}$, which is finally given to the actuators. The dynamics of the system will make the joints move and the sensors will provide the new feedback joint velocities and positions.

In the rest of these notes, we will present the fundamentals underlying each of the KCL blocks. The details on the dynamic level are provided in a separate course.

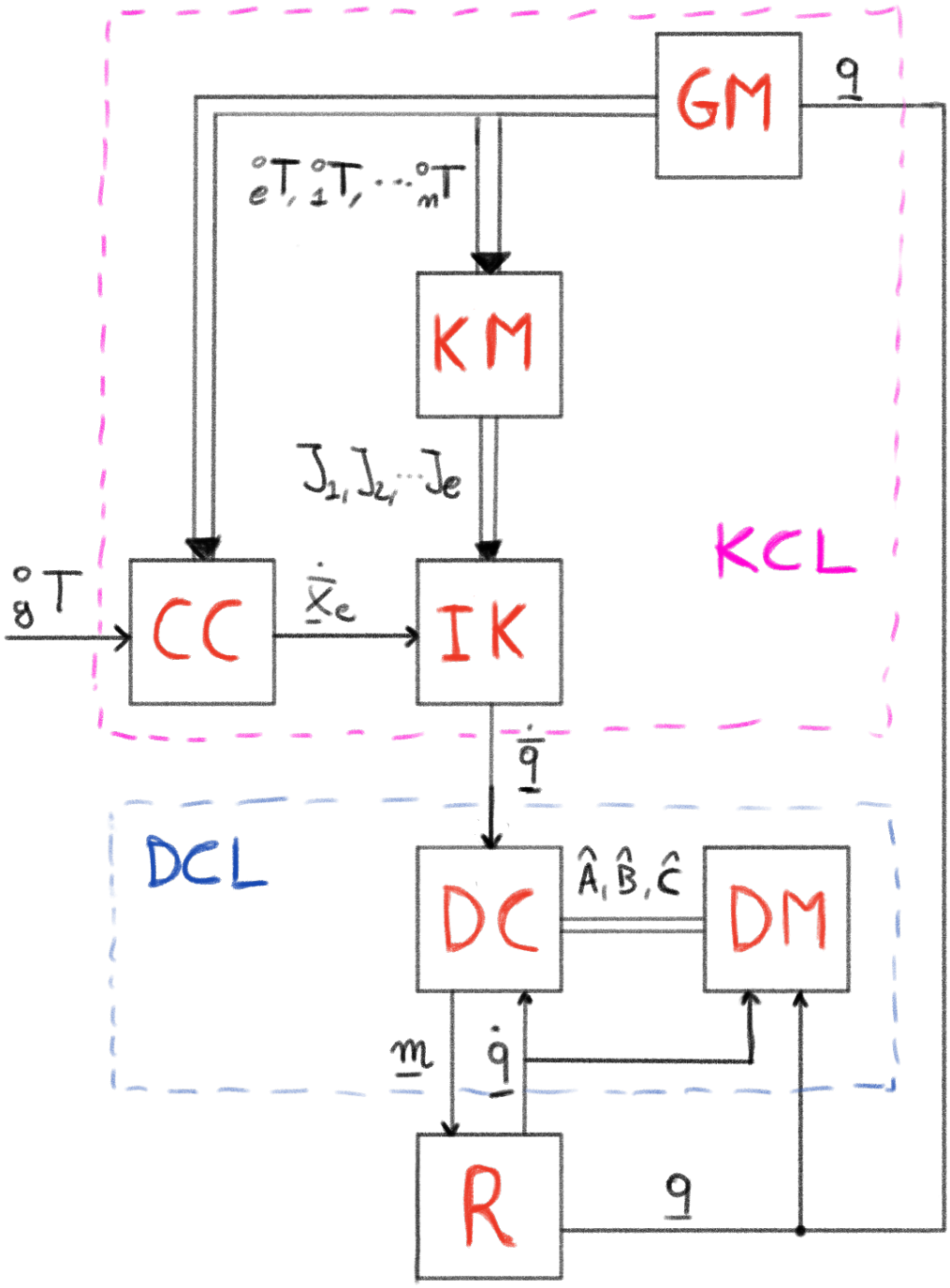


Figure 1.10: A block representation of.

Chapter 2

Geometric Fundamentals

The present chapter will review all basic geometric concepts and related algorithmic tools needed for later managing geometric problems regarding any type of robotic structure.

For sure, many of the concepts and formalisms presented in the initial sections are well known to the reader, possibly from previous courses. Even in this case, the reading of the chapter is recommended, at least for getting acquainted with the used notations.

2.1 World Coordinate System and Related Objects

A so-called *world coordinate system*, otherwise termed as the *world frame* $\langle 0 \rangle$, is represented by the *ordered triple* of *orthogonal* and *oriented axes* $x - y - z$, corresponding to a so-called right frame (i.e. with the orientation of the z axis obtained via the application of the "right-hand-rule" from x to y) that *ideal observers* have *agreed* to use for parameterizing the surrounding space.

The concept of "ideal observer" corresponds to an ideal entity capable of monitoring the evolution of the involved geometric or physical phenomena in whatever needed details, and capable of recording them with no storage limits, as well as capable of instantaneously communicating with other ideal observers via infinite capacity and speed channels. In the practice, instead, "non-ideal (real) observers" typically correspond to hardware/software subsystem organizations that, possibly distributed within the environment, perform measurements, data acquisition, data processing, data logging, data transmission, etc., but with obvious practical limitations with respect to the ideal ones.

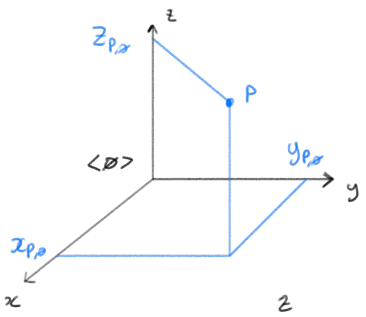


Figure 2.1: Point P and its components on frame $\langle 0 \rangle$.

With that in mind, as indicated in Fig. 2.1, the position of a point P (we shall always use capital letters for indicating points in space) with respect to the agreed frame $\langle 0 \rangle$ is termed as *point P projected on frame* $\langle 0 \rangle$ and is denoted as ${}^0P$, and is represented by the algebraic-vector of its ordered components, respectively on the $x - y - z$ axes of $\langle 0 \rangle$, that is

$$ {}^0P \triangleq \begin{bmatrix} x_{P,0} \\ y_{P,0} \\ z_{P,0} \end{bmatrix} , \tag{2.1} $$

where the indexes indicating the point P and the frame $\langle 0 \rangle$, in correspondence of each component of ${}^0P$ itself, only serve for recalling how such components are defined, and will be omitted in the rest of the notes, whenever the context is clearly understood. Moreover, let us trivially note that the position ${}^0O_0$ of the origin of frame $\langle 0 \rangle$, with respect to frame $\langle 0 \rangle$ itself, corresponds to the null position (i.e. the null algebraic vector).

Within such a parameterized space, other than points, we also have the so-called *applied projected vectors* on frame $\langle 0 \rangle$, simply defined and denoted as

$$ {}^0[P - Q] \triangleq \left[{}^0P - {}^0Q \right] , \tag{2.2} $$

corresponding to the algebraic vector of the components on $\langle 0 \rangle$ of the oriented segment that starting from point Q reaches point P.

Accordingly with its definition, an applied projected vector cannot be parallel shifted. Indeed, if we were instead parallel shifting it into another ${}^0[P' - Q']$, then we should consider it as a *different* vector. Finally, let us note how a projected point ${}^0P$ can be seen as an applied projected vector on frame $\langle 0 \rangle$, with starting point the origin of frame $\langle 0 \rangle$:

$$ {}^0P = {}^0[P - O_0]\,. \tag{2.3} $$

An important extension is the concept of *projected vector* on frame $\langle 0 \rangle$ denoted as

$$ {}^0(P - Q) \triangleq \left({}^0P - {}^0Q\right), \tag{2.4} $$

which now includes the *entire class* of applied projected vectors on frame $\langle 0 \rangle$, which are each one a parallel shift of a given seminal one ${}^0[P - Q]$. The use of round brackets, in lieu of squared ones, underlines this difference. Moreover, let us introduce a more compact vector notation:

$$ {}^0\boldsymbol{v}_{P,Q} \triangleq \left({}^0P - {}^0Q\right), \tag{2.5} $$

where the indication of the seminal points is maintained. Whenever the context is clear, the projected vector will be indicated simply as ${}^0\boldsymbol{v}$. Moreover, accordingly with the so-called *Grassman rule*, note that a position point ${}^0P$ can also be represented as

$$ {}^0P = {}^0Q + {}^0\boldsymbol{v}, \tag{2.6} $$

i.e. as a position point translated toward the desired position, accordingly with the translation provided by a given projected vector.

Finally, within the space parameterized by $\langle 0 \rangle$, a special ordered triple of projected vectors can be further introduced as

$$ \begin{aligned} {}^0\boldsymbol{i}_0 &\triangleq \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^\top \\ {}^0\boldsymbol{j}_0 &\triangleq \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^\top \\ {}^0\boldsymbol{k}_0 &\triangleq \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^\top \end{aligned} \tag{2.7} $$

corresponding to the well known so-called projected unit vectors of the x-y-z axes of frame $\langle 0 \rangle$, projected on frame $\langle 0 \rangle$ itself.

2.2 Other Coordinate Systems

Once the ideal observers have agreed about the world coordinate system $\langle 0 \rangle$, they can also agree on the use of other coordinated systems, of which at least one of them is positioned with respect to the originally agreed frame $\langle 0 \rangle$; and each one of the others still positioned with respect to $\langle 0 \rangle$ (directly or indirectly through other frames).

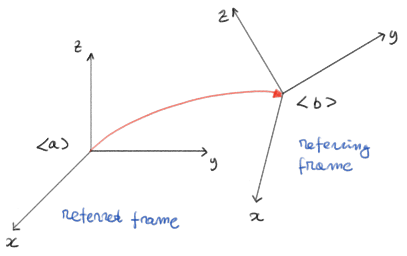


Figure 2.2: Point P and its components on frame $\langle 0 \rangle$.

Fig. 2.2 is an example of a graph indicating how a possible positioning of different frames $\langle 1 \rangle$, $\langle 2 \rangle$, $\langle 3 \rangle$ has been progressively performed by starting from the seminal one $\langle 0 \rangle$.

Within such a graph, the tips of the arrows joining one frame with another are used for indicating a corresponding positioned frame, otherwise termed as a *referring frame*, while the tails indicate the corresponding *referred frame*, i.e. the one with respect to which the frame at the tip has been positioned. By proceeding in this way a so-called tree of frames (a connected graph) can consequently be constructed, as agreed by all the observers, termed as the *agreed universe of frames*. Let us note how, when positioning a referring frame $\langle b \rangle$ with respect to a referred one $\langle a \rangle$, then the point of view of an observer fixed with the referred frame $\langle a \rangle$ (i.e. just looking to where actually is located with respect to him) is de-facto implicitly adopted.

To describe the positioning of a generic frame $\langle b \rangle$ with respect to another one $\langle a \rangle$, the following set of information data is needed:

$$ \left\{ {}^aO_b; ({}^a\boldsymbol{i}_b, {}^a\boldsymbol{j}_b, {}^a\boldsymbol{k}_b) \right\}, \tag{2.8} $$

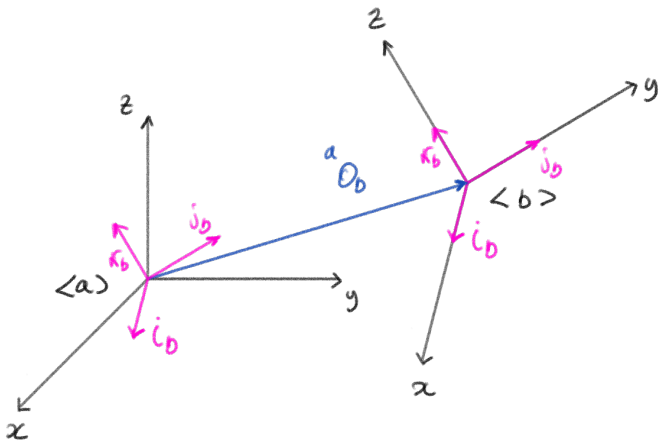


Figure 2.3: Point P and its components on frame $\langle 0 \rangle$.

composed by the where the origin O_b of frame $\langle b \rangle$ is located w.r.t. $\langle a \rangle$, i.e. the projection of O_b on the referred frame $\langle a \rangle$, indicated as ${}^aO_b$; and the ordered triple of unit vectors of $\langle b \rangle$, each one represented with components on frame $\langle a \rangle$. The second set of information serves for establishing which is the *orientation* of frame $\langle b \rangle$ with respect to frame $\langle a \rangle$.

2.2.1 Orientation Matrices

Let us now consider two frames $\langle a \rangle$, $\langle b \rangle$, and let us further consider a generic geometric vector $\boldsymbol{v}$, whose projection ${}^b\boldsymbol{v}$ is it also known. Let us now perform a change of projection coordinate for vector $\boldsymbol{v}$, i.e. let us devise a formula for deducing its projection ${}^a\boldsymbol{v}$ on frame $\langle a \rangle$. To this aim, let us formerly *express* the geometric vector $\boldsymbol{v}$ as the linear combination of the unit vectors of frame $\langle b \rangle$:

$$\boldsymbol{v} = \boldsymbol{i}_b x_b + \boldsymbol{j}_b y_b + \boldsymbol{k}_b z_b. \tag{2.9}$$

After projecting on frame $\langle a \rangle$ one obtains

$${}^a\boldsymbol{v} = {}^a\boldsymbol{i}_b x_b + {}^a\boldsymbol{j}_b y_b + {}^a\boldsymbol{k}_b z_b, \tag{2.10}$$

which rearranged in matrix form becomes

$${}^a\boldsymbol{v} = \begin{bmatrix} {}^a\boldsymbol{i}_b & {}^a\boldsymbol{j}_b & {}^a\boldsymbol{k}_b \end{bmatrix} \begin{bmatrix} x_b \\ y_b \\ z_b \end{bmatrix}. \tag{2.11}$$

Then, the 3×3 matrix

$${}^a_b\boldsymbol{R} \triangleq \begin{bmatrix} {}^a\boldsymbol{i}_b & {}^a\boldsymbol{j}_b & {}^a\boldsymbol{k}_b \end{bmatrix}. \tag{2.12}$$

is called *orientation matrix* or *rotation matrix*, and it allows to express the formula for the change of coordinates of a projected vector, i.e.

$${}^a\boldsymbol{v} = {}^a_b\boldsymbol{R}\,{}^b\boldsymbol{v}, \tag{2.13}$$

2.2.1.1 Properties of Orientation Matrices

A first noteworthy property of ${}^a_b\boldsymbol{R}$ is that, calling r_{ij} its generic entry,

$$r_{ij} \in [-1, 1] \;\; \forall \;\; i, j \in \{1, 2, 3\}. \tag{2.14}$$

Moreover, given the orthonormal nature of the basis vectors that compose an orientation matrix, it turns out orientation matrices are orthonormal, hence they satisfy

$${}^a_b\boldsymbol{R}\,{}^a_b\boldsymbol{R}^\top = \boldsymbol{I}_{3\times 3} \tag{2.15}$$

$${}^a_b\boldsymbol{R}^\top\,{}^a_b\boldsymbol{R} = \boldsymbol{I}_{3\times 3} \tag{2.16}$$

implying that

$$ {}^a_b\boldsymbol{R}^\top = {}^a_b\boldsymbol{R}^{-1}. \tag{2.17} $$

Moreover, consider again (2.13) and multiply by the inverse of the rotation matrix

$$ {}^b\boldsymbol{v} = {}^a_b\boldsymbol{R}^{-1}{}^a\boldsymbol{v} = {}^a_b\boldsymbol{R}^\top{}^a\boldsymbol{v}. \tag{2.18} $$

However, if the change of coordinate problem had been casted with the roles of frames $\langle a \rangle$ and $\langle b \rangle$ swapped, we would have obtained

$$ {}^b\boldsymbol{v} = {}^b_a\boldsymbol{R}{}^a\boldsymbol{v}, \tag{2.19} $$

then, due to the arbitrariness of $\boldsymbol{v}$, it follows that we must necessarily have that

$$ {}^a_b\boldsymbol{R}^\top = {}^b_a\boldsymbol{R}, \tag{2.20} $$

which represents the simple formula for knowing the orientation of a first frame with respect to a second one, once the orientation of the second one is known with respect to the first. In particular, the above equation implies the following inner structure of ${}^a_b\boldsymbol{R}$

$$ {}^a_b\boldsymbol{R} = \begin{bmatrix} {}^a\boldsymbol{i}_b & {}^a\boldsymbol{j}_b & {}^a\boldsymbol{k}_b \end{bmatrix} = \begin{bmatrix} {}^b\boldsymbol{i}_a^\top \\ {}^b\boldsymbol{j}_a^\top \\ {}^b\boldsymbol{k}_a^\top \end{bmatrix}, \tag{2.21} $$

which provides a very easy mean for immediately looking at the mutual orientations between two given frames.

Recalling that for any two square non singular matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ of the same size:

$$ \det(\boldsymbol{A}\boldsymbol{B}) = \det \boldsymbol{A} \det \boldsymbol{B} \tag{2.22} $$

(Binet Theorem), and that

$$ \det \boldsymbol{A} = \det \boldsymbol{A}^\top, \tag{2.23} $$

it follows that for any $3D$ matrix $\boldsymbol{M}$ for which $\boldsymbol{M}\,\boldsymbol{M}^\top = \boldsymbol{I}_{3\times 3}$ it must also be

$$ \det \boldsymbol{M} = \pm 1. \tag{2.24} $$

It follows that any $3D$ rotation matrix $\boldsymbol{R}$ must have $|\det \boldsymbol{R}| = 1$: actually, given that a rigid rotation must preserve relative orientations (i.e. a right hand side orthonormal frame must be transformed in another right hand side orthonormal frame) it follows that

$$ \det \boldsymbol{R} = +1. \tag{2.25} $$

2.2.1.2 Group Properties of Rotation Matrices in $\Re^n$

Following the same line of reasoning described before for $3D$ rotations, one can define a rotation matrix in $\Re^n$ as a matrix $\boldsymbol{R} \in \Re^{n\times n}$ such that: $\boldsymbol{R}^\top\boldsymbol{R} = I_{n\times n}$ (assuring *isometry* or preservation of the euclidean norm) and $\det \boldsymbol{R} = +1$ (assuring the preservation of relative direction). Indeed, the set of all possible $n-$dimensional matrices having such properties can be shown to form a group in $\Re^n$. In particular, define

$$ SO(n) = \left\{ \boldsymbol{R} \in \Re^{n\times n} : \boldsymbol{R}^\top\boldsymbol{R} = \boldsymbol{I}_{n\times n},\ \det \boldsymbol{R} = +1 \right\} \tag{2.26} $$

that is called the special orthogonal group where the term special refers to the fact that $\det \boldsymbol{R} = +1$ rather than $\det \boldsymbol{R} = \pm 1$. The fact that $SO(n)$ is named a group refers to the fact that its elements satisfy all the group axioms under the operation of matrix multiplication [**?**]:

- Closure: given any two elements $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ of $SO(n)$ also $\boldsymbol{R}_1\boldsymbol{R}_2$ belongs to $SO(n)$.
- Identity: there exists an element $\boldsymbol{I}$ of $SO(n)$ such that $\boldsymbol{I}\boldsymbol{R} \in SO(n)$ for any $\boldsymbol{R} \in SO(n)$.
- Inverse: for any $\boldsymbol{R} \in SO(n)$ there exists $\boldsymbol{R}^{-1} \in SO(n)$ such that $\boldsymbol{R}\boldsymbol{R}^{-1} = \boldsymbol{I}$.
- Associativity: given any three elements $\boldsymbol{R}_1, \boldsymbol{R}_2, \boldsymbol{R}_3$ in $SO(n)$ it follows that $\boldsymbol{R}_1(\boldsymbol{R}_2\boldsymbol{R}_3) = (\boldsymbol{R}_1\boldsymbol{R}_2)\boldsymbol{R}_3$.

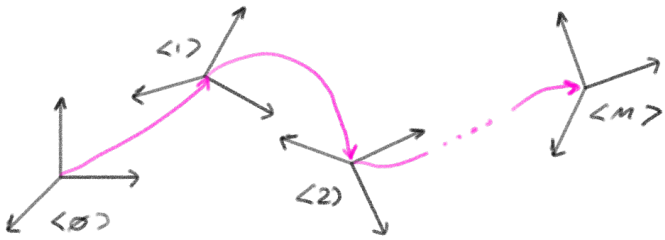


Figure 2.4: A chain of frames. By recursively going backwards the chain, each time applying the change of projection of a vector, one obtains the closure property of $SO(n)$ matrices.

The above results can be proven by direct calculation and are left as exercise. The closure property is particularly interesting in $3D$ applications as it reveals how rotations may be combined. To this aim, let us consider a chain of $n+1$ frames, as depicted in Fig. 2.4, for which we only evidence their cascaded orientation matrices ${}^{i-1}_{i}\boldsymbol{R}, i = 1, \ldots, n$. We want to deduce the orientation matrix ${}^{0}_{n}\boldsymbol{R}$ from the last frame with respect to the first one. To this aim, let us extend the problem considered in the previous section, and let us assume to know the projection ${}^{n}\boldsymbol{v}$ of a generic vector $\boldsymbol{v}$ on the last frame, and then let us evaluate its projection ${}^{0}\boldsymbol{v}$ on the first one. Repeating backwardly the change of coordinate formula developed in the previous section, one obtains:

$$ {}^{0}\boldsymbol{v} = \left({}^{0}_{1}\boldsymbol{R}\,{}^{1}_{2}\boldsymbol{R} \cdots {}^{i-1}_{i}\boldsymbol{R} \cdots {}^{n-1}_{n}\boldsymbol{R}\right) {}^{n}\boldsymbol{v}, \tag{2.27} $$

that, accordingly with the arbitrariness of $\boldsymbol{v}$, allows stating that

$$ {}^{0}_{n}\boldsymbol{R} = {}^{0}_{1}\boldsymbol{R}\,{}^{1}_{2}\boldsymbol{R} \cdots {}^{i-1}_{i}\boldsymbol{R} \cdots {}^{n-1}_{n}\boldsymbol{R}. \tag{2.28} $$

2.3 Transformation Matrices

In the previous sections we have focused on the representation of the mutual orientation between two frames, neglecting the position information represented by ${}^{a}O_b$. Let us now introduce a compact notation that takes into account both the positioning information and the orientation one in a 4×4 matrix

$$ {}^{a}_{b}\boldsymbol{T} \triangleq \begin{bmatrix} {}^{a}_{b}\boldsymbol{R} & {}^{a}O_b \\ \boldsymbol{0}_{3\times 1} & 1 \end{bmatrix}, \tag{2.29} $$

which is termed *Transformation Matrix* (sometimes also called *Homogeneous Matrix*) of frame $\langle b \rangle$ with respect to frame $\langle a \rangle$. The reasons why the above matrix is constructed to be 4×4, with the first three elements of the fourth row always zero, and the last one always unitary, will be clarified in the sequel.

Let us consider a frame $\langle b \rangle$, positioned with respect to $\langle a \rangle$ within an agreed universe, via its transformation matrix ${}^{a}_{b}\boldsymbol{T}$. Let us further assume to know the position ${}^{b}P$ of a generic point P with respect to frame $\langle b \rangle$ and let us compute its position ${}^{a}P$ with respect to frame $\langle a \rangle$.

The above stated problem can be trivially solved, by observing that it will necessarily be

$$ {}^{a}P = {}^{a}O_b + {}^{a}_{b}\boldsymbol{R}\,{}^{b}P \tag{2.30} $$

as changing the coordinates of projected point from one frame to another one always requires its reference origin to be changed accordingly, as Fig. **??** shows.

Let us now represent projected points via the so called *homogeneous coordinates*, i.e. via the following 4×1 algebraic vector

$$ {}^{b}\bar{P} = \begin{bmatrix} {}^{b}P \\ 1 \end{bmatrix}, \tag{2.31} $$

where the scalar value 1 is always added as the fourth component. Then, exploiting this representation, the change of coordinates formula for a point can be more compactly rewritten as

$$ {}^{b}\bar{P} = {}^{b}_{a}\boldsymbol{T}\,{}^{a}\bar{P}. \tag{2.32} $$

Following similar steps as done for orientation matrices, it is possible to show that

$$ {}^a_b\boldsymbol{T}^{-1} = {}^b_a\boldsymbol{T}. \tag{2.33} $$

Moreover, the inversion of a transformation matrix can be found as:

$$ {}^a_b\boldsymbol{T}^{-1} = \begin{bmatrix} {}^a_b\boldsymbol{R}^\top & -{}^a_b\boldsymbol{R}^\top {}^aO_b \\ \boldsymbol{0}_{3\times 1} & 1 \end{bmatrix}. \tag{2.34} $$

2.3.1 Evaluation of Transformation Matrices within Trees of Frames

Let us again consider a chain of $n+1$ frames, as depicted in Fig. **??**. Following the same idea exploited for orientation matrices, consider as known the position of a point P with respect to the last frame, i.e. ${}^n\bar{P}$. Then, its position w.r.t. the first frame $\langle 0 \rangle$ can be found by repeatedly using the change of coordinates formula

$$ {}^0\bar{P} = \left({}^0_1\boldsymbol{T}\,{}^1_2\boldsymbol{T} \cdots {}^{i-1}_i\boldsymbol{T} \cdots {}^{n-1}_n\boldsymbol{T}\right) {}^n\bar{P}, \tag{2.35} $$

that, accordingly with the arbitrariness of P, allows stating that

$$ {}^0_n\boldsymbol{T} = {}^0_1\boldsymbol{T}\,{}^1_2\boldsymbol{T} \cdots {}^{i-1}_i\boldsymbol{T} \cdots {}^{n-1}_n\boldsymbol{T}. \tag{2.36} $$

2.4 Free from Coordinate Notation

In the previous sections, we have shown how to change the projection of points, vectors and applied vectors to a different frame within an agreed universe of frames. It was shown how a frame can be described via different equivalent positioning with respect to the other frames. This makes the original agreed world frame $\langle 0 \rangle$ actually loosing its character of privilege, which was only apparently assigned to it.

Therefore, let us introduce the so-called *free from coordinate* notation, for indicating points, vectors and applied vectors without mentioning any projection frame since, in force of existing transformation equivalence, it can be any one among the agreed ones.

More precisely, in terms of free from coordinate notation, points, vectors, and applied vectors, will be more simply denoted as

- P - Capital letters for points;
- $\boldsymbol{v} = (P - Q)$ - bold lower case letters, or round parentheses difference of points, for vectors;
- $[P - Q]$ - square bracketed difference of points, or a vector-point couple for applied vectors;

where the introduced notation identifies, for each item, the entire class of the associated projections, which transform each other via coordinate changes.

It must be explicitly noted how, with the use of such a notation, the corresponding items result into *symbolic* items, each one of them generally termed as a *geometric-vector*, or a *geometric-point*. Instead, their projected equivalent forms are *algebraic-vectors* (i.e. 3×1 matrices, or 4×1 in case of homogeneous coordinates).

Finally note how even the notation for denoting frames, say for instance a generic frame $\langle a \rangle$, can be it also interpreted as a free-from-coordinates notation, where the explicit indication of its referring frame is avoided, since it could actually be any one of the other frames within the agreed universe.

2.5 Universes of Frames and Reference Systems

The present subsection is more of conceptual nature than technical. It will fix some further concepts (mainly the concept of Reference System) that will permeate the contents of all the successive chapters.

We can start the discussion by noticing how the agreed universe of frames (i.e. coordinate systems) as well as the Euclidean space (i.e. the whole set of points parameterized by each frame of the agreed universe) has been de-facto presented as if they were "static entities" not subjected to any change after their establishment. Of course, this is definitely not true, since if we now explicitly consider the time, and also assume the frames of a universe maintaining their "individuality along time", then we

certainly have situations where at least some of the universe frames are changing their transformation matrices along time, with respect to at least one of the other frames of the universe itself.

So, if we have a frame $\langle\alpha\rangle$ of the universe exhibiting such a behavior, at a first glance we might consequently say that is therefore "moving" inside the universe, with respect to each one of the frames with respect to which it is changing its transformation matrices.

However, if we conversely consider one or more of such other frames (i.e. those with respect to which has been at a first glance considered "moving") we could say that each one of them are moving with respect to $\langle\alpha\rangle$. This in total accordance with the trivial well known fact that the concept of motion is a *relative* one.

As the consequence of the motion relativity, when at least a frame of a universe changes in time its transformation matrix with respect to other frames, we can only say that it is the configuration of the entire universe which is actually changing. The interpretation for such changes is let to the "adopted point of view", i.e. the frame from which such changes are observed.

For a given frame $\langle\alpha\rangle$, we can consequently adopt the free-from-coordinates notation $\langle\alpha\rangle(t)$ for representing it within the time varying context induced by its supporting entire class of now time varying transformation matrices ${}^{i}_{\alpha}\boldsymbol{T}$, $i = 1, \ldots, n$, with respect to all other frames of the universe. If we also consider points characterized by their own "individuality" along time, we can use the free-from-coordinate notation $P(t)$ to denote them. A point that does not change its position with respect to a frame $\langle\alpha\rangle(t)$ is a pointed "attached" to $\langle\alpha\rangle(t)$; nevertheless, it can still be a point moving within the universe, if and only if $\langle\alpha\rangle(t)$ is moving in the previously specified relative sense.

Motions within the universe are also allowed for "individual vectors" (applied or not), as the results of the motion assigned to their identifying points.

At this point, let us consider the set of individual points which are not varying their position with respect to $\langle\alpha\rangle(t)$ itself, and let us term such a rigid space as the *Reference System* established by frame $\langle\alpha\rangle(t)$, to be for instance denote as $\Sigma_\alpha(t)$.

Let us note that any other frame $\langle b\rangle(t)$ exhibiting a constant position and orientation with respect to $\langle\alpha\rangle(t)$, shares with it the same rigid space. Hence, a reference system can also be intended as established by the entire class of additional frames that maintain constant position and orientation with respect to the given one.

Intrinsic to the definition of reference system (or rigid space) is the possibility of enlarging the original universe of frames with the infinite continuous sets of frames belonging, each one of them, to the reference systems of the original ones. This possibility makes the universe becoming continuous also with respect to frames, as it was already as regard points, vectors and applied vectors. Furthermore, the originally agreed *discrete set* of frames actually lose its character of privilege. Indeed, any collection of frames within the different reference systems could be used for constructing (or in a sense reconstructing) the same universe.

Finally, let us explicitly note how there exists a general tendency to consider the three distinct terms "frames" (i.e. "coordinate systems"), "reference systems/rigid spaces" and "ideal observers", as synonymous one of the other, notwithstanding the fact that they are slightly differently originated concepts. For instance, an observer should be more properly assigned to a reference system rather than to a specific frame, i.e. it should be delocalized within the assigned reference system. However, its attachment to a specific frame is in any case accepted, within the obvious understanding of considering it equivalent to any other attached to a frame of the same reference system (i.e. they all are de-facto the same observer).

2.6 Vector Operations

Let us now introduce how vector operations can be performed. A vector can be *expressed* on a frame, say $\langle a\rangle$, yielding

$$\boldsymbol{v} = x_a\boldsymbol{i}_a + y_a\boldsymbol{j}_a + z_a\boldsymbol{v}_a, \tag{2.37}$$

which is a symbolic representation of $\boldsymbol{v}$, i.e. still a geometric one, not an algebraic one. If we instead consider the *projection* on frame $\langle a\rangle$ we mean the following algebraic vector:

$${}^{a}\boldsymbol{v} = \begin{bmatrix} x_a \\ y_a \\ z_a \end{bmatrix}. \tag{2.38}$$

All vector operations must be projected on the same frame, and the result will also be on that frame. Consider the geometric expression

$$2\boldsymbol{v} = (x_a \boldsymbol{i}_a + y_a \boldsymbol{j}_a + z_a \boldsymbol{v}_a) + (x_b \boldsymbol{i}_b + y_b \boldsymbol{j}_b + z_b \boldsymbol{v}_b)\,. \tag{2.39}$$

In order to compute it, we need to project it on a frame. Of course, we cannot compute ${}^a\boldsymbol{v} + {}^b\boldsymbol{v}$. Instead, the correct formulation would require the use of an orientation matrix, namely

$$2{}^a\boldsymbol{v} = {}^a\boldsymbol{v} + {}^a_b\boldsymbol{R}\,{}^b\boldsymbol{v} \tag{2.40}$$

2.6.1 Linear Combinations

In general, a linear combination of n vectors, named $\boldsymbol{v}_1, \dots \boldsymbol{v}_n$ can be written in geometric, free from coordinate notation as

$$\boldsymbol{v} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + \boldsymbol{v}_n \tag{2.41}$$

Once projected on a given frame, say $\langle 0 \rangle$, the above formula becomes

$${}^0\boldsymbol{v} = c_1\,{}^0\boldsymbol{v}_1 + c_2\,{}^0\boldsymbol{v}_2 + \cdots + {}^0\boldsymbol{v}_n \tag{2.42}$$

2.6.2 Scalar Product

In free from coordinate notation, the scalar product between two vectors $\boldsymbol{v}_1, \boldsymbol{v}_2$ is defined as

$$\boldsymbol{v}_1 \bullet \boldsymbol{v}_2 = \boldsymbol{v}_2 \bullet \boldsymbol{v}_1 = \|\boldsymbol{v}_1\|\|\boldsymbol{v}_2\| \cos\theta. \tag{2.43}$$

Note that the above equation is directly defined without require any concept of frame, but it is instead simply rooted in its geometrical meaning. Once projected on a frame, say $\langle a \rangle$, the above formula becomes

$${}^a\left(\boldsymbol{v}_1 \bullet \boldsymbol{v}_2\right) = {}^a\boldsymbol{v}_1^\top\,{}^a\boldsymbol{v}_2. \tag{2.44}$$

It is then interesting to define the following linear operator

$$(\boldsymbol{v}_1 \bullet) \tag{2.45}$$

as the 1×3 matrix that performs the scalar product between the input vector and $\boldsymbol{v}_1$. Once projected on a frame, the operator simply becomes the transpose of $\boldsymbol{v}_1$ computed on that frame.

2.6.3 Vector Product

In free from coordinate notation, the vector product (also called cross product) between two vectors $\boldsymbol{v}_1, \boldsymbol{v}_2$ is defined as

$$\boldsymbol{v}_1 \times \boldsymbol{v}_2 = \begin{cases} \|\boldsymbol{v}_1 \times \boldsymbol{v}_2\| = \|\boldsymbol{v}_1\|\|\boldsymbol{v}_2\| \sin\theta \\ \boldsymbol{v}_1 \times \boldsymbol{v}_2 \perp \boldsymbol{v}_1, \boldsymbol{v}_2 \end{cases}, \tag{2.46}$$

and where the direction of the resulting vector is established according to the right hand rule. Again, the above equation is directly defined without the need of any concept of frames. The vector product is anti-commutative, i.e.,

$$(\boldsymbol{v}_1 \times \boldsymbol{v}_2) = -(\boldsymbol{v}_2 \times \boldsymbol{v}_1) \tag{2.47}$$

Once projected on a frame, say $\langle a \rangle$, the above formula becomes

$${}^a\left(\boldsymbol{v}_1 \times \boldsymbol{v}_2\right) = {}^a\left[\boldsymbol{v}_1\times\right]\,{}^a\boldsymbol{v}_2, \tag{2.48}$$

where

$${}^a\left[\boldsymbol{v}_1\times\right] \triangleq \begin{bmatrix} 0 & -z_a & y_a \\ z_a & 0 & -x_a \\ -y_a & x_a & 0 \end{bmatrix} \tag{2.49}$$

and where x_a, y_a, z_a are the components of $\boldsymbol{v}_1$ once projected on $\langle a \rangle$. Similarly to the scalar product, the linear operator

$$[\boldsymbol{v}_1\times] \tag{2.50}$$

is the 3×3 matrix that performs the vector product between the input vector and $\boldsymbol{v}_1$. The operator is also called the skew-symmetric operator, due to its anti-symmetric property.

Note that given any skew-symmetric 3×3 real matrix A there exists a unique vector $\boldsymbol{a} \in \mathbb{R}^{3\times 1}$ (called the *axial vector* associated to A) such that for any $\boldsymbol{x} \in \mathbb{R}^{3\times 1}$:

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{a} \times \boldsymbol{x}. \tag{2.51}$$

The operator associating to a skew symmetric 3×3 real matrix its unique axial vector is occasionally denoted as vex, namely

$$\begin{aligned} &\operatorname{vex}\left([\boldsymbol{a}\times]\right) = \boldsymbol{a} \\ &\boldsymbol{A}\boldsymbol{x} = \operatorname{vex}\left(\boldsymbol{A}\right) \times \boldsymbol{x} \quad \forall\, \boldsymbol{x} \in \mathbb{R}^{3\times 1},\ \forall\, \boldsymbol{A} \in \mathbb{R}^{3\times 3}\ :\ \boldsymbol{A} = -\boldsymbol{A}^{\top}. \end{aligned} \tag{2.52}$$

2.6.4 Similarity Transformation

Let us consider a $\langle b \rangle$ and any projected vector ${}^{b}\boldsymbol{v}$. Let us consider the linear operator $\boldsymbol{L} \in \mathbb{R}^{3\times 3}$ such that

$${}^{b}\boldsymbol{w} = {}^{b}\boldsymbol{L}\,{}^{b}\boldsymbol{v}. \tag{2.53}$$

Then, let us consider $\boldsymbol{w}$ projected onto another frame $\langle a \rangle$ and consider its dependency on $\boldsymbol{v}$ as it follows

$${}^{a}\boldsymbol{w} = {}^{a}_{b}\boldsymbol{R}\,{}^{b}\boldsymbol{w} = {}^{a}_{b}\boldsymbol{R}\,{}^{b}\boldsymbol{L}\,{}^{b}\boldsymbol{v} = {}^{a}_{b}\boldsymbol{R}\,{}^{b}\boldsymbol{L}\,{}^{a}_{b}\boldsymbol{R}^{\top}\,{}^{a}\boldsymbol{v} \triangleq {}^{a}\boldsymbol{L}\,{}^{a}\boldsymbol{v}. \tag{2.54}$$

The transformation ${}^{a}\boldsymbol{L} \to {}^{a}_{b}\boldsymbol{R}\,{}^{b}\boldsymbol{L}\,{}^{a}_{b}\boldsymbol{R}^{\top}$ is called a similarity transformation and the matrices ${}^{a}\boldsymbol{L}$ and ${}^{b}\boldsymbol{L}$ are called similar. Two matrices are similar if and only if they represent the same linear map under two possibly different bases, with $\boldsymbol{R}$ being the change-of-basis matrix.

If we apply the similarity transformation concept to the vector product operator we have that

$$\begin{aligned} {}^{b}\left(\boldsymbol{w} \times \boldsymbol{v}\right) &= {}^{b}\left[\boldsymbol{w}\times\right]{}^{b}\boldsymbol{v} \\ {}^{a}\left(\boldsymbol{w} \times \boldsymbol{v}\right) &= {}^{a}_{b}\boldsymbol{R}\,{}^{b}\left[\boldsymbol{w}\times\right]{}^{b}\left[\boldsymbol{v}\times\right]{}^{a}_{b}\boldsymbol{R}^{\top} \\ &= {}^{a}\left[\boldsymbol{w}\times\right]{}^{a}\boldsymbol{v} \end{aligned} \tag{2.55}$$

2.7 Other Useful Properties of Real Matrices

2.7.1 Powers of a Skew-Symmetric 3×3 Real Matrix Associated With a Unit Vector

Consider a unit vector $\boldsymbol{h} \in \mathbb{R}^{3\times 1}$. According to the definition in equation (**??**) of the associated vector product skew-symmetric matrix $[\boldsymbol{h}\times]$, the powers of such matrix (in the specific case that $\|\boldsymbol{h}\| = 1$) satisfy the following:

$$\begin{aligned} \left[\boldsymbol{h}\times\right] &= \boldsymbol{h}\times \\ \left[\boldsymbol{h}\times\right]^{2}\boldsymbol{x} &= \left[\boldsymbol{h}\times\right]\left[\boldsymbol{h}\times\right]\boldsymbol{x} = \boldsymbol{h}\times\left(\boldsymbol{h}\times\boldsymbol{x}\right) = \left(\boldsymbol{h}\boldsymbol{h}^{\top} - I_{3\times3}\right)\boldsymbol{x} \\ \left[\boldsymbol{h}\times\right]^{2} &= \left(\boldsymbol{h}\boldsymbol{h}^{\top} - I_{3\times3}\right) \\ \left[\boldsymbol{h}\times\right]^{3}\boldsymbol{x} &= -\left[\boldsymbol{h}\times\right]\boldsymbol{x} \\ \left[\boldsymbol{h}\times\right]^{3} &= -\left[\boldsymbol{h}\times\right] \\ \left[\boldsymbol{h}\times\right]^{4} &= -\left[\boldsymbol{h}\times\right]^{2} \\ \left[\boldsymbol{h}\times\right]^{5} &= -\left[\boldsymbol{h}\times\right]^{3} = \left[\boldsymbol{h}\times\right] \\ \left[\boldsymbol{h}\times\right]^{6} &= \left[\boldsymbol{h}\times\right]^{2} \\ \left[\boldsymbol{h}\times\right]^{7} &= -\left[\boldsymbol{h}\times\right] \\ &\vdots \end{aligned} \tag{2.56}$$

In general the odd and even powers of $[\boldsymbol{h}\times]$ satisfy for any nonnegative integer i:

$$\begin{aligned} \left[\boldsymbol{h}\times\right]^{2i+1} &= (-1)^{i}\left[\boldsymbol{h}\times\right] \quad \text{skew-symmetric matrix} \\ \left[\boldsymbol{h}\times\right]^{2(i+1)} &= (-1)^{i}\left(\boldsymbol{h}\boldsymbol{h}^{\top} - I_{3\times3}\right) \quad \text{symmetric matrix.} \end{aligned} \tag{2.57}$$

Notice that $-[\boldsymbol{h}\times]^2$ is the projection operator on the plane normal to the unit vector $\boldsymbol{h}$ and in particular $[\boldsymbol{h}\times]^2\,\boldsymbol{h} = \boldsymbol{0}$. Also notice that

$$\begin{aligned} \mathrm{tr}\left([\boldsymbol{h}\times]^{2i+1}\right) &= 0 \\ \mathrm{tr}\left([\boldsymbol{h}\times]^{2(i+1)}\right) &= ((-1)^{(i+1)})\,2 \end{aligned} \tag{2.58}$$

being tr the trace operator (i.e. the sum of the diagonal elements of a square matrix).

2.7.2 Decomposition in Symmetric and Skey-Symmetrical Parts

A square real matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is said to be symmetrical if $\boldsymbol{A}^\top = \boldsymbol{A}$ and is said to be skew-symmetrical if $\boldsymbol{A}^\top = -\boldsymbol{A}$. Not all matrices are either symmetrical or skew-symmetrical, yet all square real matrices may be uniquely decomposed in the sum a symmetrical and a skew-symmetrical term. Indeed for any $\boldsymbol{A} \in \Re^{n\times n}$ one has:

$$\boldsymbol{A} = \frac{\boldsymbol{A}+\boldsymbol{A}^\top}{2} + \frac{\boldsymbol{A}-\boldsymbol{A}^\top}{2} \tag{2.59}$$

where

$$\begin{aligned} \frac{\boldsymbol{A}+\boldsymbol{A}^\top}{2} &\text{ is } \quad \text{symmetric} \\ \frac{\boldsymbol{A}-\boldsymbol{A}^\top}{2} &\text{ is } \quad \text{skew-symmetric} \end{aligned} \tag{2.60}$$

2.8 Three-parameter Representation of Orientation Matrices

For the subsequent developments, it is now important to introduce other more compact ways of representing an orientation matrix ${}^a_b\boldsymbol{R}$ between two generic frames $\langle a\rangle$, $\langle b\rangle$.

To this aim, let us recall that orientation matrices are orthonormal, therefore their 9 elements must satisfy the following 6 constraints:

- Unimodularity. Its three columns (rows) are unit vectors, that is

$$\begin{aligned} {}^a\boldsymbol{i}_b^\top\,{}^a\boldsymbol{i}_b &= 1, \\ {}^a\boldsymbol{j}_b^\top\,{}^a\boldsymbol{j}_b &= 1, \\ {}^a\boldsymbol{k}_b^\top\,{}^a\boldsymbol{k}_b &= 1. \end{aligned} \tag{2.61}$$

- Orthogonality. Its three columns (rows) are orthogonal to each other, that is

$$\begin{aligned} {}^a\boldsymbol{i}_b^\top\,{}^a\boldsymbol{j}_b &= 0, \\ {}^a\boldsymbol{j}_b^\top\,{}^a\boldsymbol{k}_b &= 0, \\ {}^a\boldsymbol{k}_b^\top\,{}^a\boldsymbol{i}_b &= 0. \end{aligned} \tag{2.62}$$

Therefore, due to the existence of the 6 constraints shown above, it is possible to represent ${}^a_b\boldsymbol{R}$ with three parameters only, even if in a not unique manner.

2.8.1 Exponential Representation of Rotation Matrices in $\mathbb{R}^3$

A particularly interesting minimal representation of $3D$ rotation matrixes is the so called exponential or angle - axis representation. To introduce such parameterization, we first recall the basic properties of the exponential of a matrix operator. Given a fixed real square matrix $M \in \Re^{n\times n}$ and $\theta \in \Re$, the exponential matrix of θM belongs to $\Re^{n\times n}$ and is defined as

$$e^{\theta M} = \sum_{k=0}^{\infty} \frac{(\boldsymbol{M}\theta)^k}{k!} \tag{2.63}$$

that is always well defined as the sum on the right hand side of equation (2.63) can be shown to be always convergent. Some of the properties of exponential matrices are summarized below:

$$\boldsymbol{M}\, e^{\theta \boldsymbol{M}} = e^{\theta \boldsymbol{M}}\, \boldsymbol{M} \tag{2.64}$$

$$\left(e^{\theta \boldsymbol{M}}\right)^{\top} = e^{\theta \boldsymbol{M}^{\top}} \tag{2.65}$$

$$\left(e^{\theta \boldsymbol{M}}\right)^{-1} = e^{-\theta \boldsymbol{M}}. \tag{2.66}$$

With reference to equation (2.66) it should be noticed that $e^{\theta \boldsymbol{M}}$ is always full rank (hence invertible) even if $\boldsymbol{M}$ should be singular.

Coming back to the $3D$ case and to the representation of $SO(3)$, consider an arbitrary (but fixed) unit vector $\boldsymbol{h}$ and a real scalar θ. These quantities define a set of 3 independent variables such that the matrix $\boldsymbol{R}$ defined as

$$\boldsymbol{R} \triangleq e^{\theta [\boldsymbol{h}\times]} \tag{2.67}$$

belongs to $SO(3)$. This follows from the observation that

$$\left(e^{\theta [\boldsymbol{h}\times]}\right)^{\top} e^{\theta [\boldsymbol{h}\times]} = \left(e^{\theta [\boldsymbol{h}\times]^{\top}}\right) e^{\theta [\boldsymbol{h}\times]} = \left(e^{-\theta [\boldsymbol{h}\times]}\right) e^{\theta [\boldsymbol{h}\times]} = I_{3\times 3}. \tag{2.68}$$

This implies that $\det e^{\theta [\boldsymbol{h}\times]} = \pm 1$: yet noticing that $e^{\theta [\boldsymbol{h}\times]}\big|_{\theta=0} = I_{3\times 3}$ and recalling that the determinant of a matrix is a continuous function of its entries, it follows that actually $\det e^{\theta [\boldsymbol{h}\times]} = +1$. The quantities $\theta \in \Re$ and $\boldsymbol{h} \in \Re^{3\times 1} : \|\boldsymbol{h}\| = 1$ are referred to as the *angle - axis parameters* of the rotation matrix $e^{\theta S(\boldsymbol{h})}$. The vector $(\theta \boldsymbol{h}) \in \Re^{3\times 1} : \|\boldsymbol{h}\| = 1, \theta \in \Re$ is called the *rotation vector* associated to the rotation matrix $e^{\theta S(\boldsymbol{h})}$. The angle axis parameters θ and $\boldsymbol{h}$ can be grouped in a $4D$ real vector $\boldsymbol{n}u = (\boldsymbol{h}^{\top}, \theta)^{\top}$ called the *angle-axis vector (or $4D-$vector)* associated to the rotation matrix $e^{\theta S(\boldsymbol{h})}$. Notice that given $\|\boldsymbol{h}\| = 1$ the angle - axis parameters and the angle axis $4D-$vector are actually equivalent to 3 independent variables.

Lemma 1 *(surjective and non-injective nature of the exponential map to $SO(3)$)*
The mapping $e^{\theta [\boldsymbol{h}\times]} \mapsto SO(3)$ is surjective (onto), namely for any element $R \in SO(3)$ there exists multiple distinct sets of angle - axis parameters $(\boldsymbol{h}, \theta)$, multiple distinct rotation vectors $\theta \boldsymbol{h}$ and multiple distinct $4D-$angle axis vectors $\boldsymbol{\nu} = (\boldsymbol{h}^{\top}, \theta)^{\top}$ such that $\boldsymbol{R} = e^{\theta [\boldsymbol{h}\times]}$ where $\|\boldsymbol{h}\| = 1$ and $\theta \in \mathbb{R}$.

Notice that the above Lemma states an even stronger condition than non-injection. For a function to be non injective, it is sufficient to find *one* element in its range that is reached by more than one element in its domain. In the present case we can prove that *every* element is $SO(3)$ can be reached by more than one rotation vector, angle axis parameter pair or angle - axis $4D-$vector.

To prove the above Lemma it is first useful to determine an alternative formula for the exponential matrix $e^{\theta [\boldsymbol{h}\times]}$. In particular, on the basis of the power properties of $[\boldsymbol{h}\times]$ given by equations (2.57) and (**??**) and of the matrix exponential definition (2.63), it follows that:

$$\begin{aligned} e^{\theta [\boldsymbol{h}\times]} &= I_{3\times 3} + \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \ldots\right) [\boldsymbol{h}\times] + \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \ldots\right) [\boldsymbol{h}\times]^2 = \\ &= I_{3\times 3} + (\sin\theta)\, [\boldsymbol{h}\times] + (1 - \cos\theta)\, [\boldsymbol{h}\times]^2 \end{aligned} \tag{2.69}$$

that is known as the Rodrigues formula. Besides its fundamental practical importance relative to the computation of the exponential $e^{\theta [\boldsymbol{h}\times]}$, the Rodrigues formula is also a most useful tool for the interpretation of exponential and rotation matrices. Notice, in particular, that $(\sin\theta)\, [\boldsymbol{h}\times]$ is the skew-symmetric part of $e^{\theta [\boldsymbol{h}\times]}$ where as $I_{3\times 3} + (1 - \cos\theta)\, [\boldsymbol{h}\times]^2$ is its symmetrical part. Moreover, recalling the properties of the powers of $[\boldsymbol{h}\times]$ described in section **??**, the Rodrigues formula reveals that $\boldsymbol{h}$ is the unit vector of the rotation axis of the rotation matrix $e^{\theta [\boldsymbol{h}\times]}$ as $e^{\theta [\boldsymbol{h}\times]} \boldsymbol{h} = \boldsymbol{h}$. The quantity θ can be interpreted as the rotation angle around the axis $\boldsymbol{h}$. As shown in [**?**], the Rodrigues formula can be generalized to the case of the exponential of $\theta\, [\boldsymbol{v}\times]$ with $\|\boldsymbol{v}\| \neq 1$ giving rise to

$$e^{\theta [\boldsymbol{v}\times]} = I_{3\times 3} + \frac{\sin(\theta \|\boldsymbol{v}\|)}{\|\boldsymbol{v}\|}\, [\boldsymbol{v}\times] + \frac{(1 - \cos(\theta \|\boldsymbol{v}\|))}{\|\boldsymbol{v}\|^2}\, [\boldsymbol{v}\times]^2 \tag{2.70}$$

2.8.2 Equivalent Inverse Angle Axis Problem

Coming back to the proof of Lemma 1, it must be shown that given an arbitrary element $\boldsymbol{R}$ of $SO(3)$ we can determine (at least one) $\theta \in \mathbb{R}$ and $\boldsymbol{h} \in \mathbb{R}^{3\times 1} : \|\boldsymbol{h}\| = 1$ such that $\boldsymbol{R} = e^{\theta[\boldsymbol{h}\times]}$. The proof is constructive: according to the Rodrigues formula and to equation (2.58), the trace of $e^{\theta[\boldsymbol{h}\times]}$ is $1 + 2\cos\theta$, in fact

$$\begin{aligned}\mathrm{tr}(I_{3\times 3} + (\sin\theta)\,[\boldsymbol{h}\times] + (1-\cos\theta)\,[\boldsymbol{h}\times]^2) &= 3 + 0 + (1-\cos\theta)\mathrm{tr}([\boldsymbol{h}\times]^2) = 3 + (1-\cos\theta)(-2)\\ &= 1 + 2\cos\theta.\end{aligned} \tag{2.71}$$

On the other hand, given that $\boldsymbol{R} \in SO(3)$ is an isometry (i.e. it preserves the euclidean norm), its eigenvalues must have unit length and they occur in conjugate pairs [**?**]. It follows that for any $\boldsymbol{R} \in SO(3)$ the trace $\mathrm{tr}(\boldsymbol{R})$ must satisfy $-1 \leq \mathrm{tr}(R) \leq 3$. Consequently by equating the traces of $\boldsymbol{R}$ and of $e^{\theta[\boldsymbol{h}\times]}$, one gets the condition:

$$\mathrm{tr}(\boldsymbol{R}) = 1 + 2\cos\theta \implies \tag{2.72}$$

$$\theta = \cos^{-1}\left(\frac{\mathrm{tr}(\boldsymbol{R}) - 1}{2}\right) \tag{2.73}$$

that results always well defined in the light of the mentioned property $-1 \leq \mathrm{tr}(\boldsymbol{R}) \leq 3$ for any element $\boldsymbol{R} \in SO(3)$. Assuming that the $\cos^{-1}(\cdot)$ function has range $[0, \pi]$, equation (2.73) allows to associate a unique $\theta^* \in [0, \pi]$ to a given $\boldsymbol{R}^* \in SO(3)$.

2.8.2.1 Computing the axis vector

Special case $\theta^* = 0$: If $\boldsymbol{R} = I_{3\times 3}$ then $\theta^* = 0$ and the choice of $\boldsymbol{h}$ is completely arbitrary (hence there are infinite rotation vectors and angle axis $4D-$vectors giving rise to the same $\boldsymbol{R}^*$).

Special case $\theta^* = \pi$: If, instead, equation (2.73) implies $\theta^* = \pi$, then according to equation (2.69) the matrix $e^{\theta^*[\boldsymbol{h}\times]}$ is symmetric and equal to

$$e^{\pi[\boldsymbol{h}\times]} = 2\boldsymbol{h}\,\boldsymbol{h}^\top - I_{3\times 3}.$$

In this case by equating the above to the given $\boldsymbol{R}^* \in SO(3)$ the unit vector $\boldsymbol{h}$ can be computed by normalizing the eigenvector of $\boldsymbol{R}^*$ associated to the eigenvalue $+1$. Denoting with $\boldsymbol{h}^*$ the result of such operation, notice that also $-\boldsymbol{h}^*$ and $\theta^* = \pi$ would correspond to the same $\boldsymbol{R}^*$, proving also in this case the non uniqueness of the mapping from rotation vectors, angle axis $4D-$vectors and angle axis parameters to $SO(3)$. An explicit formula relating $\boldsymbol{h}^*$ to $\boldsymbol{R} \in SO(3)$ when $\theta^* = \pi$ can be derived noting that:

$$R = e^{\pi[\boldsymbol{h}\times]} = 2\boldsymbol{h}\,\boldsymbol{h}^\top - I_{3\times 3} = \begin{pmatrix} 2h_1^2 - 1 & 2h_1h_2 & 2h_1h_3 \\ 2h_1h_2 & 2h_2^2 - 1 & 2h_2h_3 \\ 2h_1h_3 & 2h_2h_3 & 2h_3^2 - 1 \end{pmatrix} \tag{2.74}$$

implying

$$|h_i| = +\sqrt{\frac{r_{ii} + 1}{2}} \quad \forall\, i = 1, 2, 3 \tag{2.75}$$

that are always well posed as the generic entry r_{ij} of $R \in SO(3)$ satisfies $r_{ij} \in [-1, 1]$ (equation (**??**)). The two solutions $\pm\boldsymbol{h}^*$ can be computed, by example, as follows:

$$\text{select one } i^\dagger \in \{1, 2, 3\} : \left|\sqrt{\frac{r_{i^\dagger\, i^\dagger} + 1}{2}}\right| > 0 \tag{2.76}$$

$$h_{i^\dagger} = \pm\sqrt{\frac{r_{i^\dagger\, i^\dagger} + 1}{2}} \tag{2.77}$$

$$h_j = \mathrm{sgn}(h_{i^\dagger})\,\mathrm{sgn}(r_{i^\dagger j})\sqrt{\frac{r_{jj} + 1}{2}} \quad \forall\, j \in \{1, 2, 3\} \setminus i^\dagger \tag{2.78}$$

being $\mathrm{sgn}(\cdot) : \Re \longrightarrow \{-1, 0, 1\}$ the sign function defined as:

$$\mathrm{sgn}(x) = \begin{cases} +1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases} . \tag{2.79}$$

The unit vector $\boldsymbol{h}$ associated to $R \in SO(3)$ may be always computed by normalizing the eigenvector of R relative to the eigenvalue $+1$. Yet if θ^* given by the right hand side of equation (2.73) is $\theta^* \neq 0$ and $\theta^* \neq \pi$, there is a computationally more efficient manner to compute $\boldsymbol{h}$. In particular, denoting with r_{ij} the (i, j) element of $R \in SO(3)$, its skew-symmetric component is given by:

$$\frac{R - R^\top}{2} = \frac{1}{2} \begin{pmatrix} 0 & r_{12} - r_{21} & r_{13} - r_{31} \\ r_{21} - r_{12} & 0 & r_{23} - r_{32} \\ r_{31} - r_{13} & r_{32} - r_{23} & 0 \end{pmatrix} \tag{2.80}$$

revealing immediately that its axial vector $\boldsymbol{a}$ is

$$\boldsymbol{a} = \frac{1}{2} \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix} . \tag{2.81}$$

On the other hand, according to the Rodrigues formula (2.69) the axial vector of of the skew symmetrical part of $e^{\theta[\boldsymbol{h}\times]}$ is $(\sin\theta)\,\boldsymbol{h}$. Hence, by equating these two axial vectors one gets:

$$(\sin\theta)\,\boldsymbol{h} = \frac{1}{2} \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix} = \mathrm{vex}\left(\frac{R - R^\top}{2} \right) \tag{2.82}$$

that allows to compute $\boldsymbol{h}$ in the stated hypothesis that $\theta \neq 0$, $\theta \neq \pi$. To conclude the proof about the non-injective nature of the exponential map of $SO(3)$, notice that according to equation (2.69) the scalar θ may be always changed to $\theta + 2n\pi$ for any integer n giving rise to the same $R \in SO(3)$. Moreover, with reference to the angle axis $4D-$vector (and to the angle axis parameters), notice that the two distinct vectors $\boldsymbol{n}u = (\boldsymbol{h}^\top, \theta)^\top$ and $-\boldsymbol{n}u = (-\boldsymbol{h}^\top, -\theta)^\top$ always identify the same element of $SO(3)$.